

2) Plantes general

$$h[m] = \delta[m] - \delta[m-4] \Rightarrow Y[m] = X[m] * h[m] \Rightarrow X[m] - X[m-4]$$

e) $X[m] = \cos(\omega_0 m T_s)$

$$Y[m] = \cos(\omega_0 m T_s) - \cos(\omega_0 (m-4) T_s)$$

(Sist LTI cos entrada ex. compleja $\Rightarrow$ salida $\Rightarrow$ $\hat{y}$ misma frecuencia ω_0 de $\hat{x}$)
 y depende de $H(e^{j\omega_0 T_s})$

Faciendo $\cos \Rightarrow H(e^{j\omega_0 T_s}) = 1 - e^{-j4\omega_0 T_s} = e^{-j2\omega_0 T_s} (e^{j2\omega_0 T_s} - e^{-j2\omega_0 T_s})$

$$\Rightarrow \frac{2j \sin(2\omega_0 T_s) e^{-j2\omega_0 T_s}}{1}$$

Usando $j = e^{j\pi/2}$:

$$H(e^{j\omega_0 T_s}) = 2 \sin(2\omega_0 T_s) e^{j(\frac{\pi}{2} - 2\omega_0 T_s)}$$

$$\hookrightarrow A = |H(e^{j\omega_0 T_s})| = 2 |\sin(2\omega_0 T_s)|$$

$$\varphi = \angle H(e^{j\omega_0 T_s}) = \begin{cases} \frac{\pi}{2} - 2\omega_0 T_s & \text{si } \sin(2\omega_0 T_s) \geq 0 \\ \frac{3\pi}{2} - 2\omega_0 T_s & \text{si } \sin(2\omega_0 T_s) < 0 \end{cases}$$

$$\Rightarrow \boxed{Y[m] = 2 |\sin(2\omega_0 T_s)| \cdot \cos(\omega_0 \cdot m \cdot T_s + \varphi)}$$

$$b) x[m] = (1/2)^m \cdot u[m]$$

$$y[m] = (1/2)^m u[m] - (1/2)^{m-4} u[m-4]$$

1. Bei $0 \leq m \leq 3$ ist der zweite Term 0, $y[m] = (1/2)^m$

2. Bei $m \geq 4$ ist $y[m] = (1/2)^m - (1/2)^{m-4} = (1/2)^{m-4} [(1/2)^4 - 1] = -\frac{15}{16} (1/2)^{m-4}$

$$y[m] = \begin{cases} (1/2)^m & 0 \leq m \leq 3 \\ -\frac{15}{16} (1/2)^{m-4} & m \geq 4 \\ 0 & \forall \text{ else} \end{cases}$$

$$c) x[m] = u[m+1] - u[m-2]$$

Es ist für $x[m] = 1$ in $m = -1, 0, 1$

Entwickeln $x[m] = \delta[m+1] + \delta[m] + \delta[m-1]$

ist $y[m] = x[m] - x[m-4]$

$$y[m] = \delta[m+1] + \delta[m] + \delta[m-1] - \delta[m-3] - \delta[m-4] - \delta[m-5]$$

$$\begin{cases} y[m] = 1 & m = (-1, 0, 1) \\ y[m] = -1 & m = (3, 4, 5) \\ y[m] = 0 & \forall m \end{cases}$$